

Programme: B E – Information Science & Engineering
Internal Assessment – II

TERM : 03/03/2025 to 21/06/2025	COURSE NAME : Database Management Systems
DATE : 17/06/2025 TIME: 3.00-4.00 pm	COURSE CODE : IS44
MAX MARKS: 30 Sem: 4	PORTIONS : L20 - L 42



Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1 a)	Consider the following relations for a database that keeps track of auto sales in a car dealership CAR(SerialNo, Model, Manufacturer, colour, Price) SALES(Salesperson-id, SerialNo, Date, Sale-Price) SALESPERSON(Salesperson-id, Name, Phone) Perform the following operations in SQL a) Retrieve the details of the car sold by the salesperson named 'Harry'. b) Retrieve the number of cars in each colour. c) Retrieve the most popular car model based on number of cars sold.	L3	CO3	5
b)	Analyze with an example, how 'Boyce-Codd normal form is found to be stricter than third normal form'.	L4	CO4	5
c)	What are the characteristics of a NoSQL database? Discuss any two types of NoSql Databases.	L2	CO5	5
2 a)	Explain how a GROUP BY clause works. State the difference between a WHERE and HAVING clause with an example.	L2	CO3	5
b)	Let $F = \{ssn \rightarrow Ename, Pnumber \rightarrow \{Pname, Plocation\}, \{ssn, Pnumber\} \rightarrow Hours\}$. Find the following closure sets with respect to F. $\{Pnumber\}^+, \{ssn, Pnumber\}^+$	L3	CO4	5
c)	Compare binary locks to exclusive /shared locks. Why is the latter type of locks preferable?	L2	CO5	5



3 a)	Consider the following relations for a DB that keeps track of employees working in the company and the cities that they live in. WORKS (Eno, Cname, Salary) LIVES (Eno, Ename, Street, City) LocatedIn (Cname, City) Manager (Eno, Mgrname) Where Eno is employee number, Ename is employee name and Cname is company name Specify Queries using SQL on the database. i. Find the names of employees who do not work for company "TRE" ii. Retrieve the number of employees working in each company. iii. List the details of employees who are managers.	L3	CO3	5
b)	Consider the relation EMP-PROJ= {SSN, Pnumber, Hours, Ename, Pname, Plocation}. Assume {SSN, Pnumber} as primary key. The dependencies are: SSN Pnumber → {Hours} SSN → {Ename} Pnumber → {Pname, Plocation} Normalize the above relation into 2NF.	L3	CO4	5
c)	Draw a state diagram, and discuss the typical states that a transition goes through during execution.	L2	CO5	5

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create